

## Multiple Choice (10 marks)

1. Which measurement would best be rounded to the nearest thousand?
  - (a) The number of yards run by a player in a game
  - (b) The number of people at a professional football game
  - (c) The length of a football in inches
  - (d) The weight of a professional football player in pounds
  - (e) The number of touchdowns in a professional football season
2. Maddie will ride her bike a total of 56 miles over 7 days. She will ride the same number of miles each day. What is the total number of miles Maddie will ride each day?
  - (a) 9
  - (b) 63
  - (c) 56
  - (d) 8
  - (e) 28
3. Let  $y = f(x)$  be a solution of the differential equation  $x \, dy + (y - xe^x) \, dx = 0$  such that  $y = 0$  when  $x = 1$ . What is the value of  $f(2)$ ?
  - (a)  $2/e$
  - (b)  $e^2 + 2$
  - (c)  $1/(2e)$
  - (d)  $e^2/2$
  - (e)  $2/(e^2)$
4. Consider the two sets  $X = \{10, 30, 45, 50, 55, 70, 90\}$  and  $Y = \{10, 30, 35, 50, 65, 70, 90\}$ . Which of the following is false?
  - (a) The sets have identical medians.
  - (b) The sets have identical means.
  - (c) The sets have identical ranges.
  - (d) None of the above are false.
5. Tara can develop 2 rolls of film in about 18 minutes. At this rate, how long will it take her to develop 8 rolls of film?
  - (a) 1 hour 44 minutes

- (b) 1 hour 20 minutes
- (c) 1 hour
- (d) 1 hour 12 minutes
- (e) 1 hour 30 minutes

## True / False (10 marks)

*Answer True or False*

6. True or False: The number 46,504, when rounded to the nearest thousand, results in 47,000.
7. True or False: If A and B are both orthogonal square matrices with  $\det A = -\det B$ , then  $\det(A + B)$  equals 0.
8. True or False: The greatest common factor of 36 and 90 is 18, which is the largest integer that divides both numbers without a remainder.
9. True or False: The length of segment ZY in kite WXYZ is 25.3 units, based on the properties of the diagonals and triangle similarity.
10. True or False: The positive square root of 200 is approximately 18% larger than the positive square root of 121.

## Long Form Response (20 marks)

*Answer each question with a clear and complete explanation.*

11. Given the equations  $x = 4 \cos(t)$  and  $y = 8 \sin(x)$ , derive the second derivative  $y''_{xx}$  and evaluate it at  $t = \frac{\pi}{3}$ . Discuss your methodology and findings.
12. Using Euler's method, analyze the iterative steps taken to approximate the solution of the differential equation  $dy/dx = y^2 e^x$  with the initial condition  $y(0) = 0.01$  at  $x = 6$  and a step size of  $h = 0.5$ . What is the value of  $y(6)$ , and discuss the implications of your findings?

*End of Paper*